



INSURANCE SCALE • MATHEMATICAL SYMMETRY • OPERATIONAL GROWTH

Law of Large Numbers + OMNIBAL AP Series

A detailed insurance framework for predictability, structured aggregation, operational capacity and portfolio growth

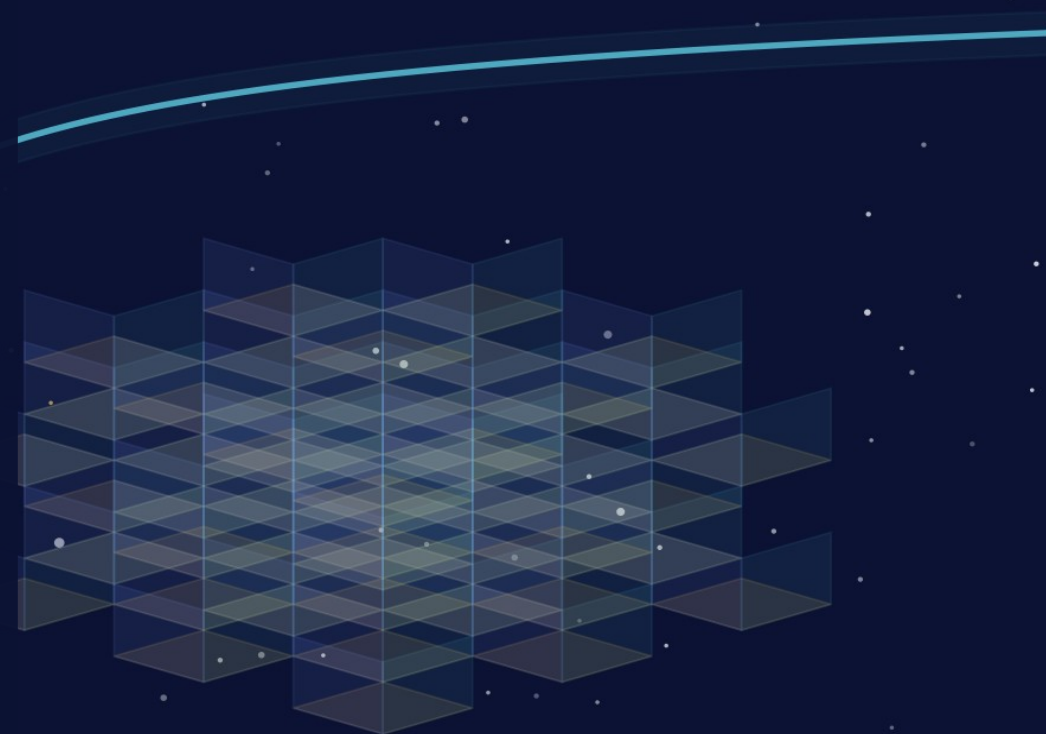
PROBABILISTIC ENGINE

DETERMINISTIC ENGINE

GROWTH LAYER

Kumaran Kanniappan • Andrometocs Research • July 2026

Research concept - not actuarial, legal, accounting or regulatory advice.



What this PDF does - and what it does not do

A bridge between insurance probability, exact power-sum mathematics and real operating models.

01 Explain LLN

Why larger, well-formed pools make average losses more stable.

02 Define OMNIBAL

How structured arithmetic-progression power sums are aggregated exactly.

03 Connect both engines

LLN estimates uncertainty; OMNIBAL accelerates eligible deterministic sums.

04 Model operating capacity

Translate claims frequency, handling time and portfolio growth into workload and FTE.

05 Set honest boundaries

Dependencies, catastrophes, arbitrary data and model governance remain essential.

06 Provide an adoption plan

A controlled path from proof-of-concept to production analytics.

The central proposition

Insurance growth needs two different kinds of scale: statistical scale and computational scale.

$$1 / \sqrt{n}$$

Relative portfolio noise

Under standard independence and finite-variance assumptions, relative uncertainty falls approximately with the square root of exposure count.

$$O(1)^*$$

Formula term count

For fixed power d , OMNIBAL uses a fixed number of algebraic terms with respect to X . Bit complexity and data preparation still matter.

$$\Sigma \rightarrow \text{model}$$

Structured aggregation

Polynomial pricing, workload or exposure functions over arithmetic-progressive cohorts can be collapsed into exact power sums.

Capacity

Operational outcome

Faster scenarios plus more stable forecasts can improve staffing, batch windows, capital planning and product growth decisions.

Most important framing

OMNIBAL does not replace the Law of Large Numbers, actuarial models, Monte Carlo simulation, GLMs, reserving standards or regulatory capital models. It is a deterministic aggregation companion for specific structured calculations.

Two engines, one insurance growth system

The strongest application is not “LLN replaced by a formula”; it is a disciplined combination.



LLN improves the quality of the estimate. OMNIBAL improves the speed of eligible aggregation. Operations converts both into business capacity.

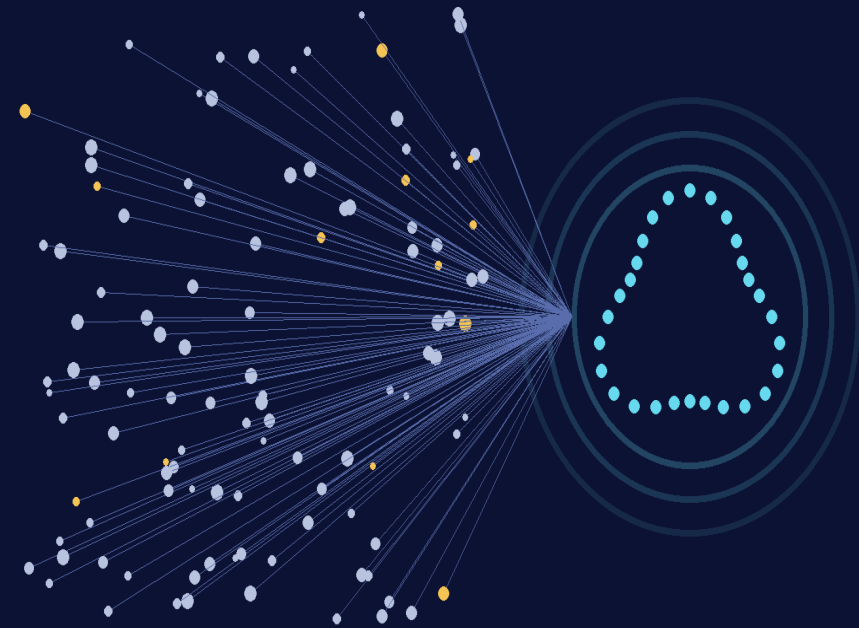
The Law of Large Numbers

As the number of suitable exposure units grows, the observed average loss tends toward the underlying expected loss.

$$\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i \rightarrow \mu$$

Insurance interpretation

- Individual claims remain uncertain.
- The average experience of a sufficiently large, well-specified pool becomes more stable.
- Pricing, reserving and staffing can therefore be based on aggregate expectations rather than prediction of each individual claim.



Many uncertain exposure units → one more predictable aggregate pool

What improves as the pool grows

For independent, identically distributed losses with finite mean and variance.

$$\mathbb{E}[S_n] = n\mu, \quad \text{Var}(S_n) = n\sigma^2, \quad \text{CV}(S_n) = \frac{\sigma}{\mu\sqrt{n}}$$

Aggregate mean

$$\mathbf{E}[S_n] = n\mu$$

Expected total loss grows linearly with the number of exposure units.

Aggregate variance

$$\mathbf{Var}(S_n) = n\sigma^2$$

Absolute variance rises because the portfolio is larger.

Relative volatility

$$\mathbf{CV}(S_n) = \sigma/(\mu\sqrt{n})$$

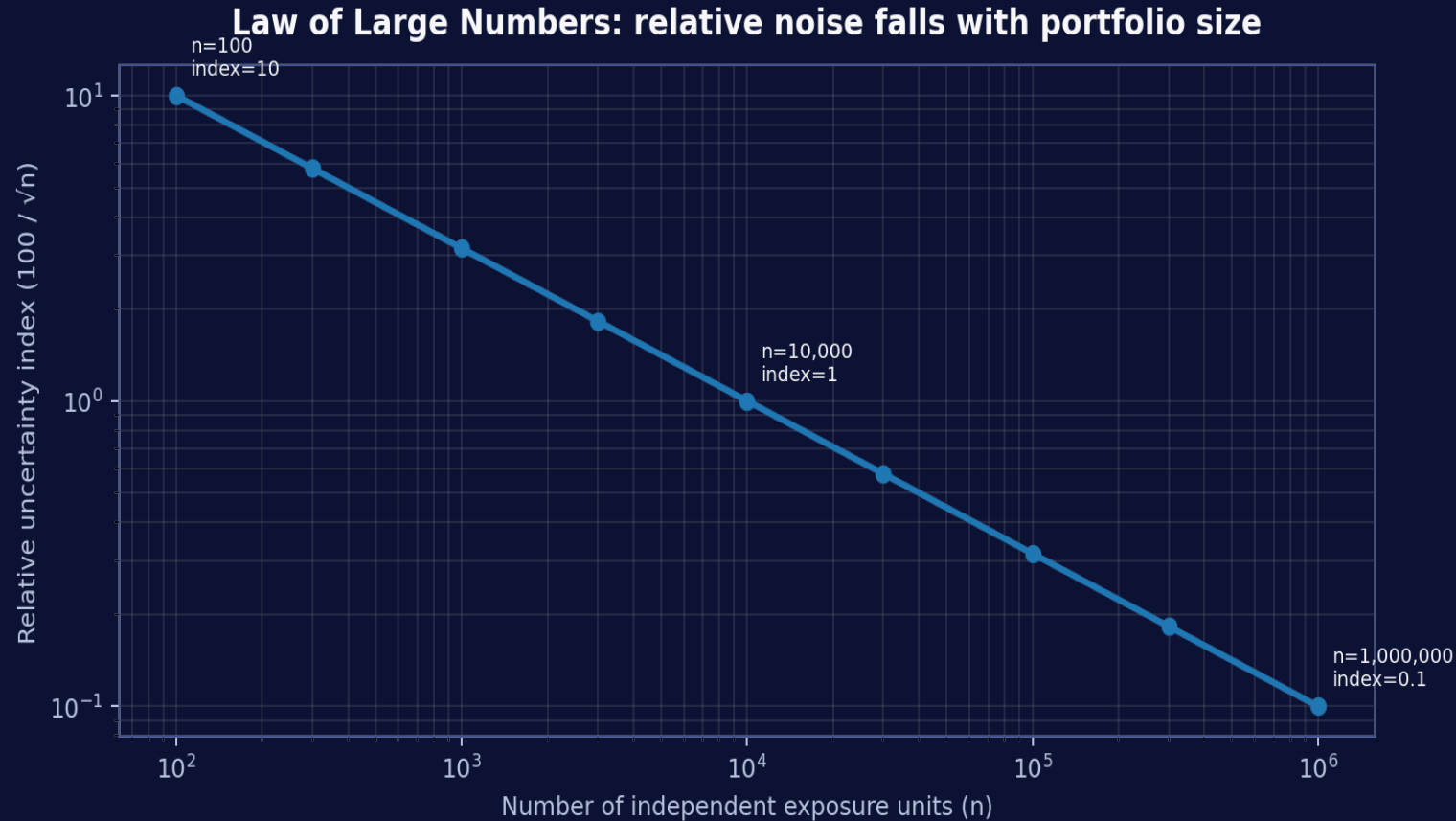
Uncertainty relative to expected loss falls with $\sqrt{n}$.

Growth rule: to cut relative statistical noise in half, a portfolio generally needs about four times as many independent exposure units.

Technical note: dependence, heterogeneity, parameter uncertainty and heavy tails can materially weaken this simple scaling.

Portfolio size creates statistical capacity

A larger pool does not eliminate losses; it makes the average experience easier to estimate.



Illustrative index based on $1/\sqrt{n}$. It is not a portfolio-specific confidence interval.

How to read the curve

- 100 exposures → uncertainty index 10
- 10,000 exposures → index 1
- 1,000,000 exposures → index 0.1
- 100× more exposure units → about 10× lower relative noise

Operational effect

More stable claim-volume forecasts support staffing, vendor capacity, liquidity, reinsurance and service-level planning.

When the Law of Large Numbers helps - and when it can fail

The size of the pool is only one requirement. Portfolio design and dependence matter.

Favourable conditions

✓	Many exposure units	Sufficient volume for credible aggregate experience.
✓	Similarity / segmentation	Risks should be grouped into reasonably homogeneous classes.
✓	Weak dependence	One loss should not automatically trigger many others.
✓	Stable process	Frequency, severity and coverage mix should not change without modeling.

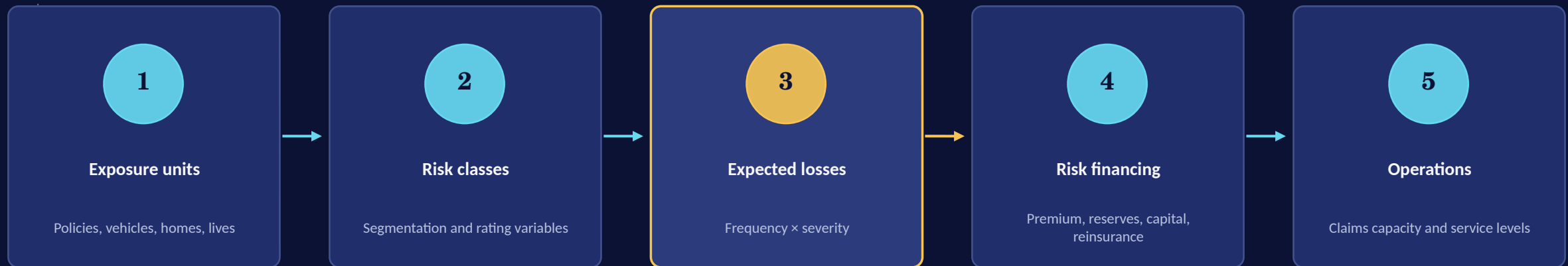
Failure modes to control

!	Catastrophe correlation	Storm, pandemic, cyber and wildfire events can hit many policies together.
!	Heavy-tailed severity	Rare extreme losses can dominate the mean and capital requirement.
!	Adverse selection	The observed pool can differ from the priced population.
!	Model and parameter risk	A precise answer to a wrong assumption is still wrong.

Sources: SOA and CAS actuarial principles; NAIC insurance-regulator materials.

How insurance converts individual uncertainty into portfolio predictability

LLN supports the aggregate estimate; capital and reinsurance absorb the remaining tail risk.



Statistical scale does not remove volatility. It makes the expected aggregate more credible, allowing capital, reinsurance and operations to be designed around a measurable distribution.

What LLN improves operationally

Better claim-volume estimates can reduce both under-capacity and over-capacity.

Staffing

Forecast FNOL, adjuster, examiner and payment workloads.

Vendor capacity

Plan repair networks, medical reviewers, counsel and catastrophe partners.

Cash and treasury

Estimate payment timing, liquidity and collateral needs.

Reinsurance

Size retentions, limits and reinstatement support.

Service levels

Protect response time and settlement turnaround as the book grows.

Product growth

Know when volume adds diversification and when it adds correlated concentration.

The next bottleneck is computation: how quickly can the insurer aggregate many structured scenarios?

OMNIBAL AP Series in insurance language

A closed-form engine for power sums over evenly spaced cohort values.

Sequence $a_k = F + (k-1)s$

Count **X cohorts or indexed values**

Midpoint $M = (F + L)/2$

Power **d controls polynomial degree**

Eligible insurance patterns

- Risk scores or sum-insured bands increasing by a fixed step
- Polynomial loadings by tier or cohort
- Structured handling-time and frequency functions
- Piecewise arithmetic-progressive approximations of a larger portfolio
- Capacity models expressed as sums of powers

Not a universal database query accelerator

Arbitrary claim records, categorical interactions, exponential-link GLMs and correlated simulations still require their own methods.

Framework basis: K. Kannappan, Cubic Sequence Formula Research Paper and OMNIBAL specialization decks, July 2026.

The OMNIBAL master formula

For $a_k = F + (k-1)s$, $L = F + (X-1)s$ and $M = (F+L)/2$.

$$D_d = \sum_{k=1}^X a_k^d = X \sum_{m=0}^{\lfloor d/2 \rfloor} \binom{d}{2m} M^{d-2m} S^{2m} \mu_{2m}(X)$$

C(d,2m)

Pascal coefficients

Even positions from row d

M^{d-2m}

Midpoint powers

Location and scale of the sequence

S^{2m}

Spacing dressing

Gap between adjacent cohort values

μ_{2m}(X)

Moment blocks

Reusable polynomials depending only on X

**Fixed number of formula terms with respect to X for a fixed d. Exact arithmetic cost still grows with the number of digits.*

Moment blocks μ_0 through μ_6

These encode the symmetric offsets around the midpoint and are reused at every power.

$$\mu_0 = 1, \quad \mu_2 = \frac{X^2 - 1}{12}, \quad \mu_4 = \frac{3X^4 - 10X^2 + 7}{240}, \quad \mu_6 = \frac{3X^6 - 21X^4 + 49X^2 - 31}{1344}$$

μ_0	1	D ¹ and every higher power
μ_2	(X ² -1)/12	D ² , D ³ and above
μ_4	(3X ⁴ -10X ² +7)/240	D ⁴ , D ⁵ and above
μ_6	(3X ⁶ -21X ⁴ +49X ² -31)/1344	D ⁶ , D ⁷ and above

Generated once by exact Bernoulli-Faulhaber arithmetic; reused as a library in structured insurance calculations.

One symmetry generates every power

The derivation is a five-step midpoint expansion.

1

Recenter

Write $a_k = M + s \cdot j_k$. Offsets j_k occur symmetrically around zero.

2

Expand

Apply the binomial theorem to $(M+s \cdot j)^d$.

3

Cancel

Odd powers of j cancel between $+j$ and $-j$.

4

Collect

Replace each average even moment by $\mu_{2m}(X)$.

5

Read off

Use Pascal coefficients to obtain any fixed power d .

The mathematical contribution is a midpoint-and-moment reformulation of classical sums of powers, not a replacement for Faulhaber's theory.

A five-band insurance score ladder

Risk scores: 10, 12, 14, 16, 18. Therefore F=10, s=2, X=5, L=18, M=14.



Step 1 - second power sum

$$\sum a_k^2 = 5[14^2 + 2^2 \cdot 2] = 1,020$$

Direct check: $100 + 144 + 196 + 256 + 324 = 1,020$

Step 2 - polynomial workload

Suppose handling units per band are

$$g(a) = 8 + 0.5a + 0.02a^2$$

$$\begin{aligned} \text{Total} &= 8X + 0.5D_1 + 0.02D_2 \\ &= 40 + 35 + 20.4 = 95.4 \text{ units} \end{aligned}$$

This is the insurance bridge: once a cost, exposure or workload function is polynomial, its total becomes a weighted combination of OMNIBAL power sums.

From actuarial variables to AP power sums

The sequence index can represent an ordered risk, exposure or operating dimension.

a_k Risk score / sum-insured band / policy-age band / development index

N_k Number of policies or claims in cohort k

$\lambda(a_k)$ Expected claim frequency per exposure

$m(a_k)$ Expected claim severity or cost

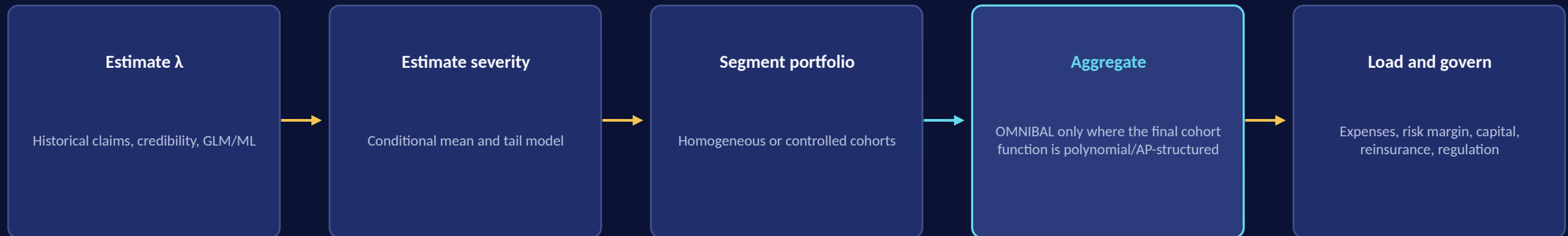
$\tau(a_k)$ Expected handling time or operational effort

If N_k , $\lambda(a_k)$, $m(a_k)$ or $\tau(a_k)$ are polynomial - or approximated piecewise by polynomials - their products can be expanded and aggregated with the required D_j values.

Expected loss aggregation

LLN supports the estimation of frequency and severity; OMNIBAL can aggregate an eligible structured expectation.

$$\mathbb{E}[L_{\text{portfolio}}] = \sum_{k=1}^X N_k \lambda(a_k) \mathbb{E}[Z \mid a_k]$$



Example eligible form: expected loss per policy $e(a) = \beta_0 + \beta_1 a + \beta_2 a^2$. Then total expected loss = $\beta_0 X + \beta_1 D_1 + \beta_2 D_2$.

OMNIBAL accelerates the deterministic sum after the actuarial assumptions are produced; it does not estimate those assumptions by itself.

Expected workload as a portfolio sum

Convert claim frequency and handling time into annual demand for adjusters, examiners and support teams.

$$W = \sum_{k=1}^X N_k \lambda(a_k) \tau(a_k), \quad FTE = \frac{W}{H_{productive}}$$

N_k

Policies in cohort k

$\lambda(a_k)$

Expected claims per policy

$\tau(a_k)$

Minutes or hours per claim

W

Total expected workload

$H_{productive}$

Annual productive hours per FTE

If $N_k \cdot \lambda(a_k) \cdot \tau(a_k)$ expands into a polynomial in a_k , the total annual workload is a weighted combination of D_0 through D^d .

Practical outputs: FTE requirement, catastrophe surge gap, outsourcing capacity, automation benefit and service-level risk.

From mathematical workload to operating decisions

The formula is useful only when connected to a measurable capacity model.

Required FTE

$$\text{FTE} = W / H_{\text{productive}}$$

Throughput capacity

$$\text{Cases} = \text{FTE} \times H_{\text{productive}} / \text{average effort}$$

Service-level buffer

$$\text{Capacity} \geq \text{expected demand} + \text{volatility buffer}$$

Scenario capacity

$$\text{Runs per batch} = \text{batch window} / \text{end-to-end run time}$$

Operational capacity increase is not automatic. It appears only if faster aggregation removes a real bottleneck and the organization acts on the improved forecast.

100,000-policy operating model

1,000 ordered risk bands × 100 policies per band. Assumptions are illustrative, not calibrated.

Claim frequency

$p_k = 0.02 + 0.00001k$

Handling time

$\tau_k = 20 + 0.02k$ minutes

Band count

$k = 1 \dots 1,000$

Policies per band

100

Expanded workload polynomial

$100 \times p_k \times \tau_k$
 $= 40 + 0.06k + 0.00002k^2$ minutes per band

Total annual workload

76,706.67 minutes
= 1,278.44 hours
≈ 0.91 FTE at 1,400 productive hours

2,501

Expected claims

Sum of band-level claim probabilities.

1.97%

Relative claim-count SD

Independent Bernoulli approximation.

D₀-D₂

OMNIBAL requirement

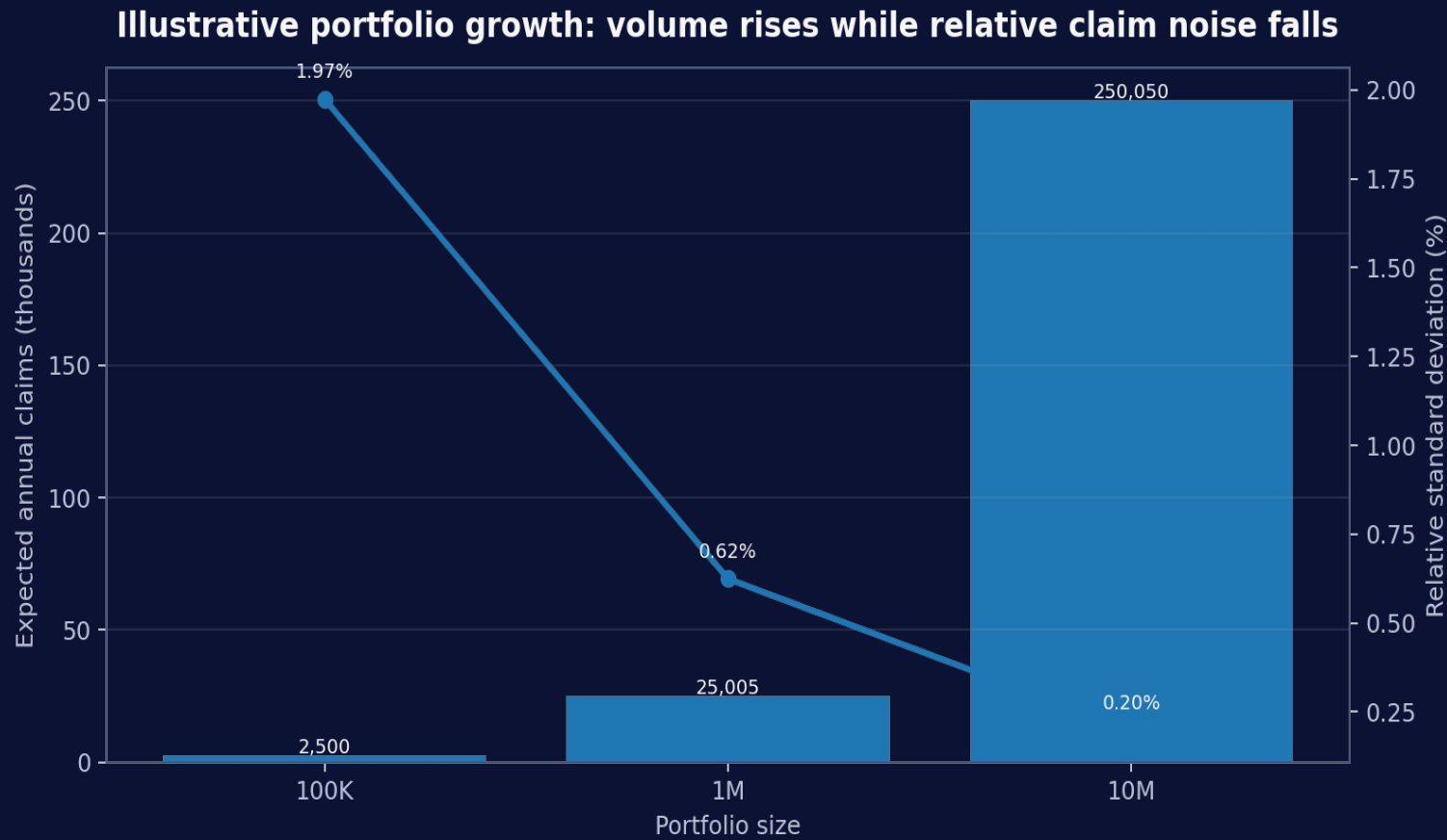
Only three power sums are needed after expansion.

The operational model can be recalculated for different portfolio sizes, automation levels or handling-time curves without iterating through every policy.

Scenario mathematics uses independent claim indicators and polynomial operating assumptions solely for demonstration.

Volume increases expected claims - but reduces relative noise

Illustrative extension of the same 1,000-band risk mix.



Growth interpretation

- 100K policies: ~2,501 expected claims; 1.97% relative SD
- 1M policies: ~25,005 expected claims; 0.62% relative SD
- 10M policies: ~250,050 expected claims; 0.20% relative SD
- Expected workload scales with volume; forecast error grows much more slowly

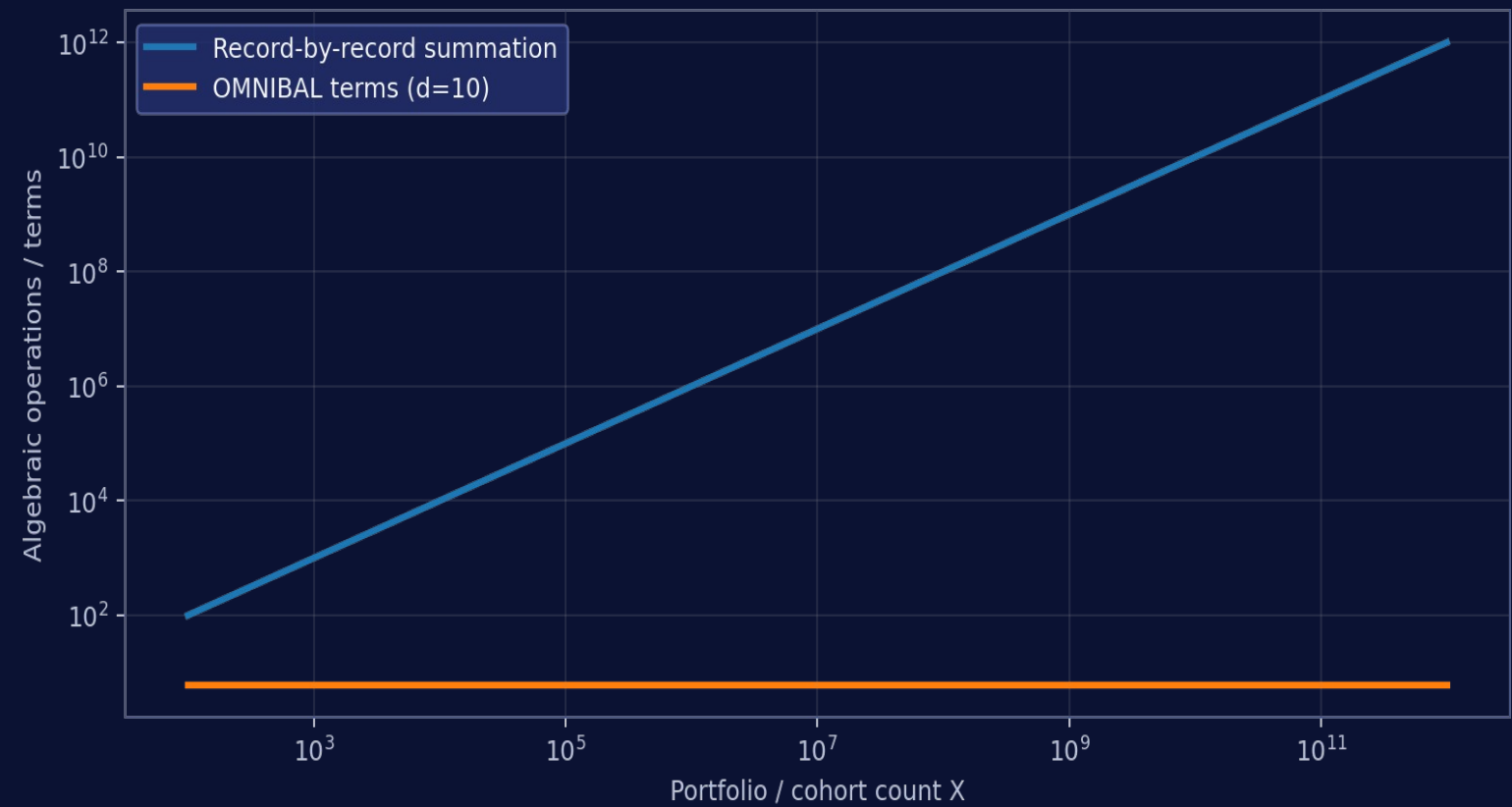
Strategic implication

A growing insurer must add operational capacity roughly with demand, while the uncertainty buffer can grow at a slower rate when risks remain well diversified.

Why structured closed forms matter

The number of formula terms is fixed for a fixed power, even when X becomes enormous.

Structured aggregation: iteration grows with X; formula term count does not



Honest complexity statement

- Fixed $d \rightarrow \text{floor}(d/2)+1$ moment terms
- Independent of X in algebraic term count
- Exact arithmetic avoids floating-point loss
- Large integers still require more bit operations
- Data extraction and cohort formation can remain $O(X)$ unless pre-aggregated

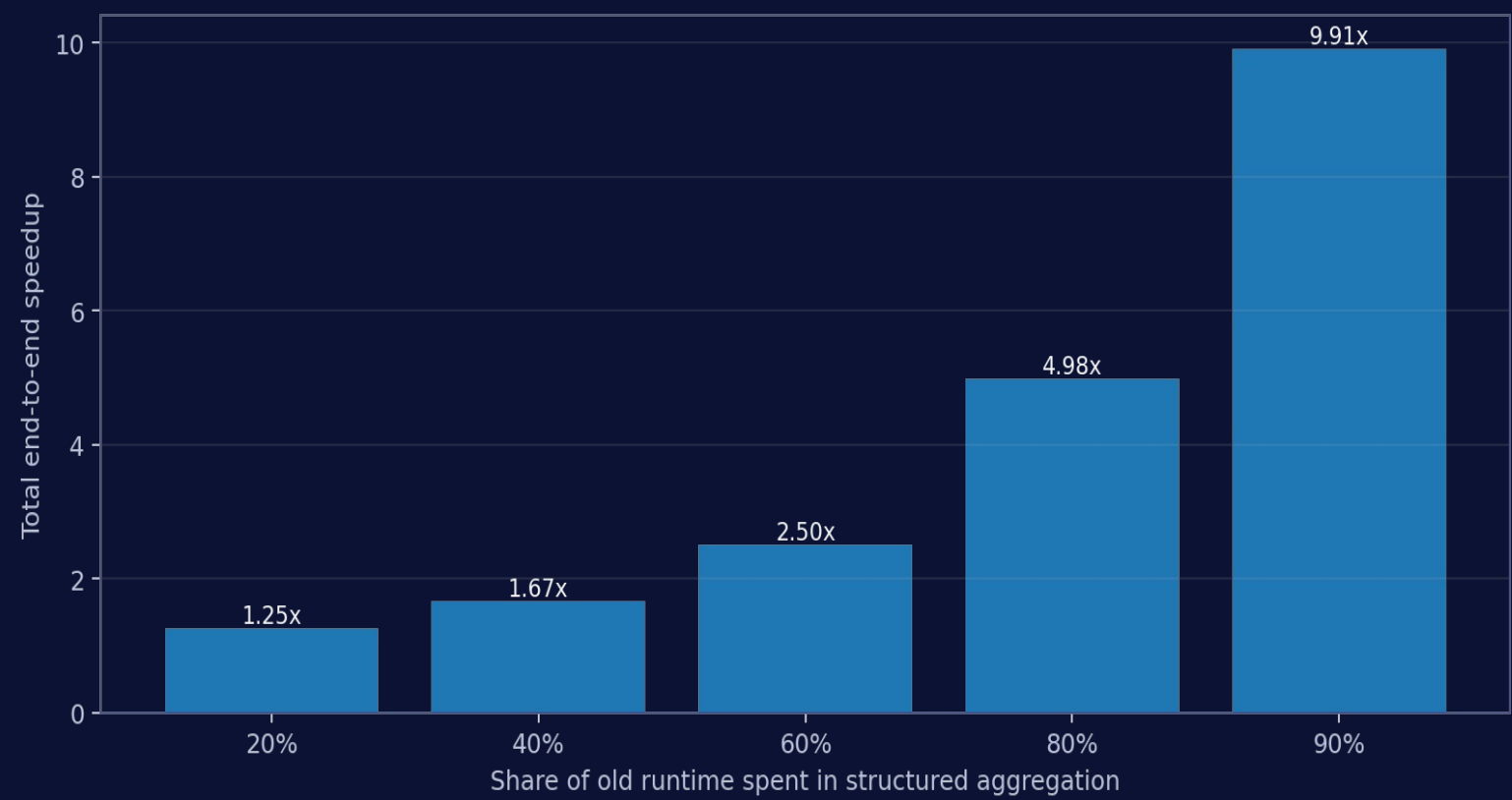
Best fit

Repeated scenarios over already structured or pre-aggregated cohorts.

End-to-end speedup follows Amdahl's law

Accelerating one component cannot remove time spent in data preparation, model scoring, controls and output.

Operational capacity gain is bounded by the rest of the workflow



Assumption

The structured aggregation component is accelerated 1,000x.

Result

If aggregation was only 20% of runtime, total capacity rises about 1.25x. If it was 80%, total capacity can approach 5x.

Measure before promising

Benchmark the actual production pipeline, including I/O, governance and integration overhead.

Pricing, exposure and product analytics

Candidate use cases where cohort values and transformations are structured.

Tiered exposure totals

Sum insured, insured value or risk score rises in fixed steps.

Polynomial risk loadings

Approved or exploratory rating functions include powers of an ordered variable.

Commission and incentive ladders

Payments depend on production bands or polynomial performance curves.

What-if pricing grids

Thousands of structured scenarios over the same cohort ladder.

Portfolio mix planning

Evaluate growth across ordered product, age or coverage tiers.

Piecewise approximations

Approximate a nonlinear curve with controlled AP-polynomial segments.

Control point: production pricing still requires actuarial soundness, discrimination review, regulatory compliance and out-of-sample validation.

Reserving, capital and reinsurance analytics

The framework can support deterministic sub-calculations, not replace the stochastic models.

Reserving

- Structured development-index sums
- Expense-allocation ladders
- Scenario aggregation over precomputed factors

Boundary

Does not replace reserving methods or professional judgment.

Capital

- Exposure aggregation by ordered risk band
- Stress-grid recalculation
- Exact totals beyond floating-point range

Boundary

Does not replace tail, dependency or regulatory capital models.

Reinsurance

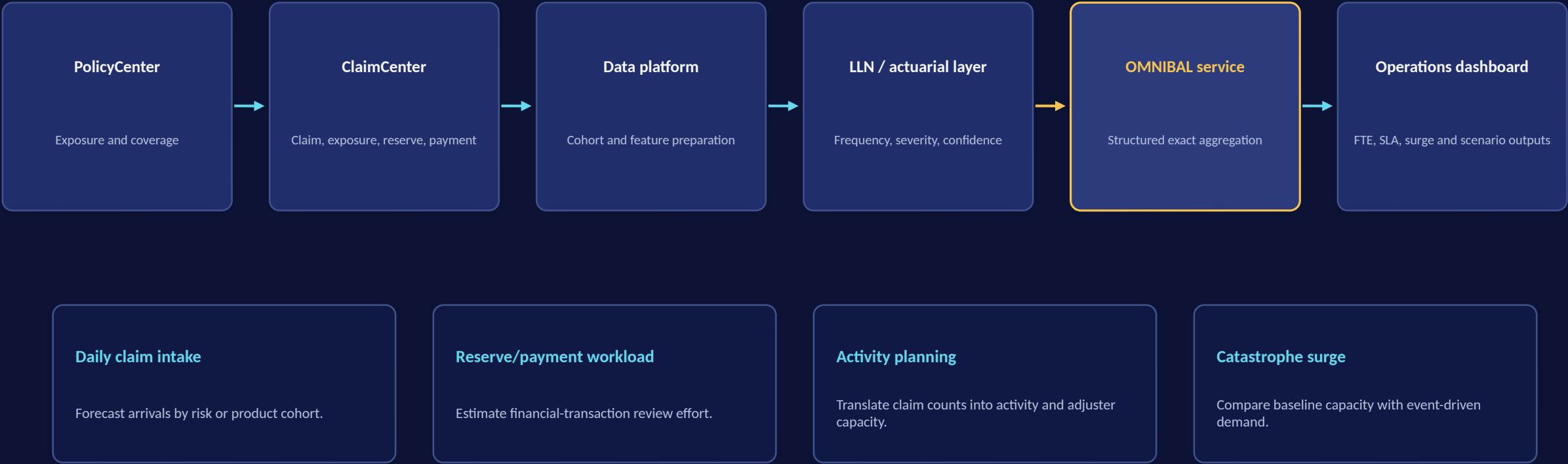
- Layer and attachment scenario grids
- Portfolio totals by retention tier
- Repeated treaty option comparisons

Boundary

Does not price treaty terms without loss distributions and contract logic.

Claims operations and Guidewire integration

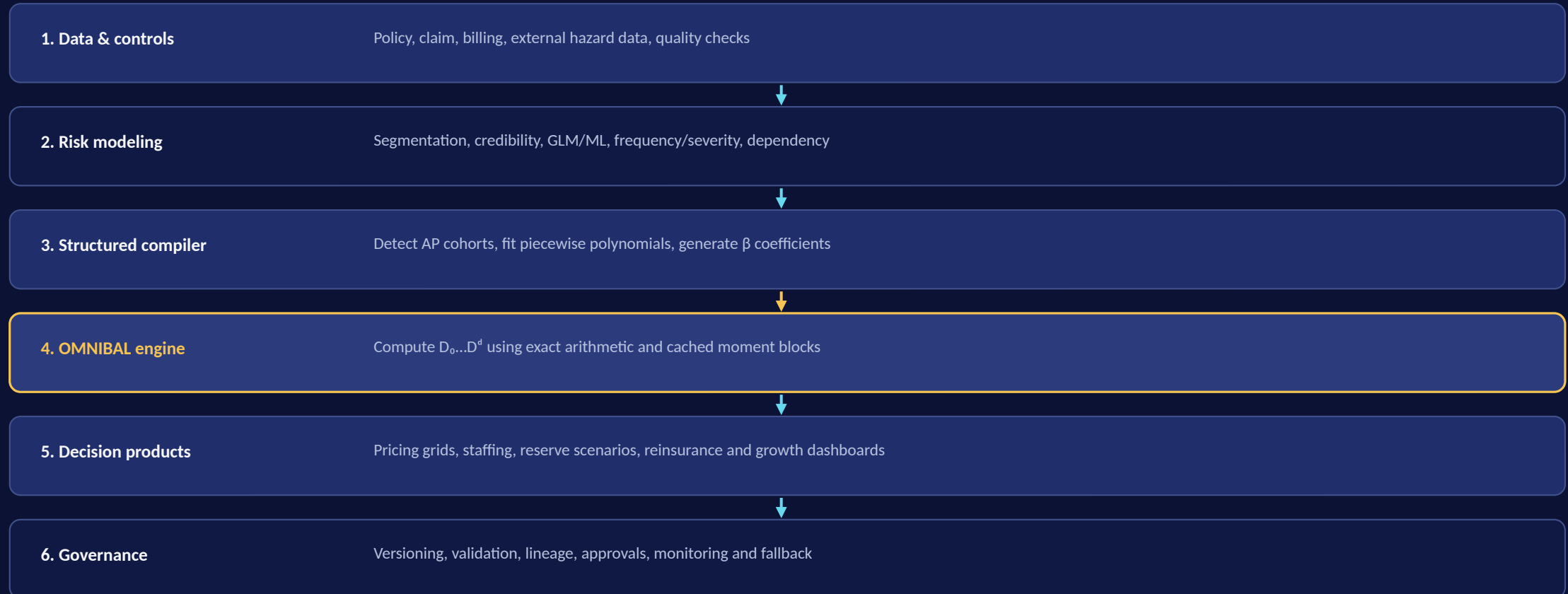
A candidate analytics service around ClaimCenter, not a replacement for ClaimCenter transaction processing.



Recommended pattern: send only controlled, pre-aggregated cohort parameters to the formula service; retain full audit lineage back to policy and claim data.

A governed insurance analytics pipeline

Separate data preparation, probabilistic estimation, deterministic aggregation and decision execution.



Fallback principle: if structure detection fails, revert to conventional aggregation rather than forcing the formula.

Validation, controls and honest boundaries

A mathematically exact formula can still be operationally wrong if the inputs or structural assumptions are wrong.

01	Structural fit	Is the cohort sequence actually arithmetic-progressive?
02	Polynomial fit	Is the target function polynomial or acceptably approximated?
03	Numerical validation	Exact match against record-level sums on representative samples.
04	Actuarial validation	Frequency, severity, dependency and tail assumptions approved.
05	Operational validation	FTE, cycle-time and service-level model back-tested.
06	Production monitoring	Drift, exceptions, fallback rate and realized-vs-expected outcomes.

Do not use the LLN as a reason to ignore concentration risk, contagion, catastrophe accumulation or changing portfolio mix.

From mathematical proof to insurance operating value

A staged program with measurable gates.



Success metrics: exactness, runtime, scenario throughput, fallback rate, forecast error, claims-per-FTE, batch-window reduction and realized business value.

A credible insurance story for OMNIBAL

Use the framework as a deterministic aggregation companion to actuarial probability - not as a substitute.

Final synthesis

- LLN: larger, well-designed pools improve predictability.
- OMNIBAL: eligible AP-polynomial aggregates can be computed exactly with a fixed formula structure.
- Operations: better forecasts and faster scenarios can increase planning capacity.
- Growth: demand rises with portfolio volume, while relative statistical noise may fall.
- Governance: dependencies, tail risk, data quality and regulatory standards remain decisive.

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